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Department of Computer Science



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Course:

Computer Network Lab - CS-241L

Reg No:

UW-24-CS-BS-101

UW-24-CS-BS-108

UW-24-CS-BS-063

Section:

BS-CS-3rd-B

CN Lab Report

Abstract

This project presents the design and implementation of a **Software House Network** for an organization named **SoftTech**. The objective of the project is to create a reliable, scalable, and secure network infrastructure that supports multiple departments and common network services such as **HTTP, FTP, SMTP, and DNS**. The network is designed using a hierarchical model with proper IP addressing, switching, routing, and service configuration. The implementation is carried out using a network simulation tool, and the results demonstrate successful communication, service accessibility, and efficient network management.

1. Introduction

In modern software houses, efficient communication and data sharing are critical for productivity and collaboration. A well-designed computer network enables developers, managers, and servers to interact seamlessly. The **SoftTech Software House Network** project focuses on designing a structured network that meets real-world organizational requirements, including centralized services, secure communication, and future scalability.

2. Problem Definition

The main problem addressed in this project is the lack of a structured and secure network for a growing software house. The organization requires:

- Centralized network services (Web, Mail, FTP)
- Reliable communication between departments
- Proper IP address management
- Secure and controlled access to network resources

Scope:

The project covers LAN design, IP addressing, service configuration, and basic security.

Constraints:

- Limited hardware resources
- Static topology

3. Literature Review

Existing enterprise network designs commonly use hierarchical architectures consisting of core, distribution, and access layers. Technologies such as **VLANs** and **client-server models** are widely used to improve performance and manageability. Previous studies highlight that centralized services reduce configuration overhead and improve scalability, but require proper planning and security mechanisms.

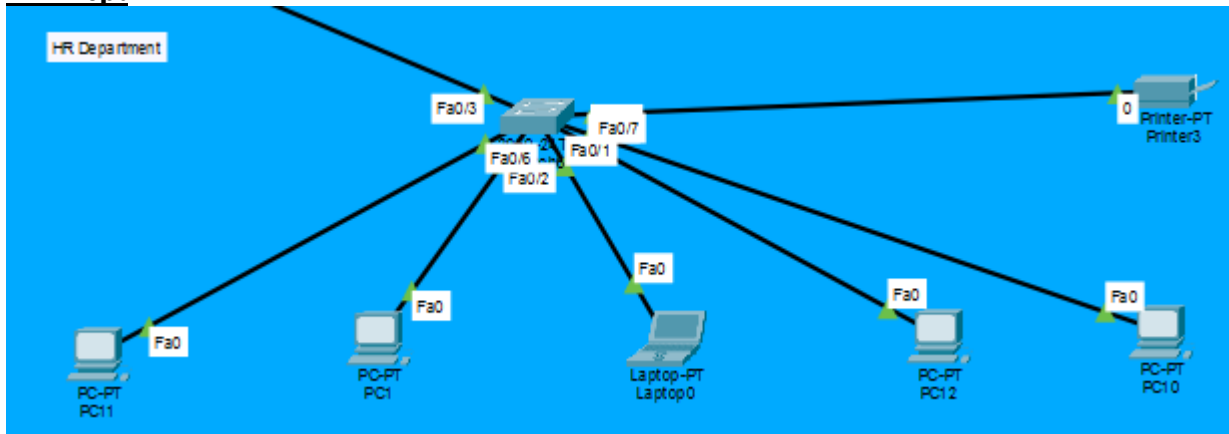
4. System Architecture

The SoftTech network follows a **client-server architecture** with a hierarchical network model:

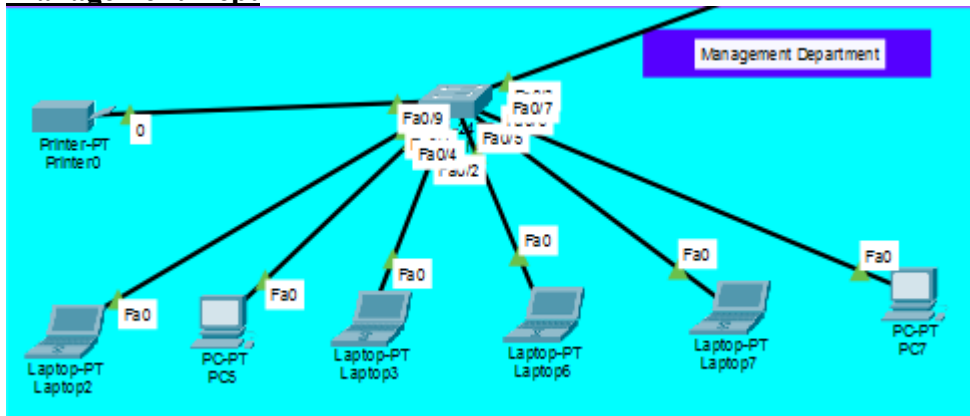
- **Core Switch:** Central backbone of the network
- **Access Switches:** Connect end devices
- **Servers:** DHCP, Web, Mail, FTP
- **End Devices:** PCs and laptops for employees
- **4 Departments:** End to end connection

The architecture ensures efficient data flow and easy network expansion.

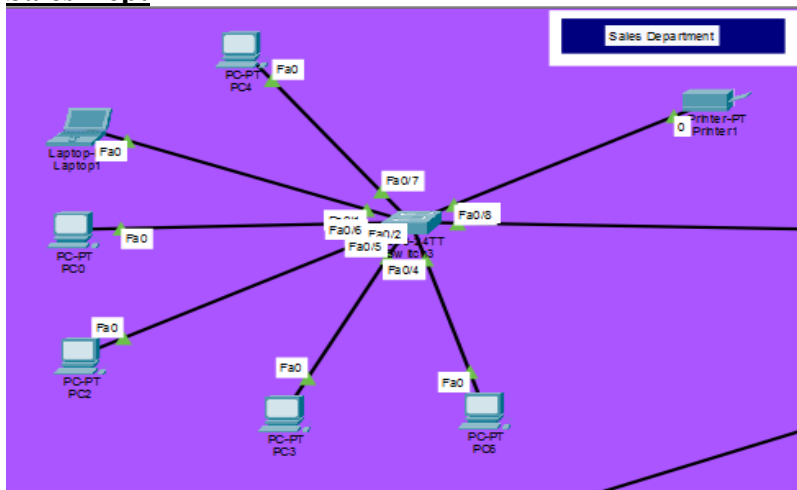
Screen Shot: HR Dept



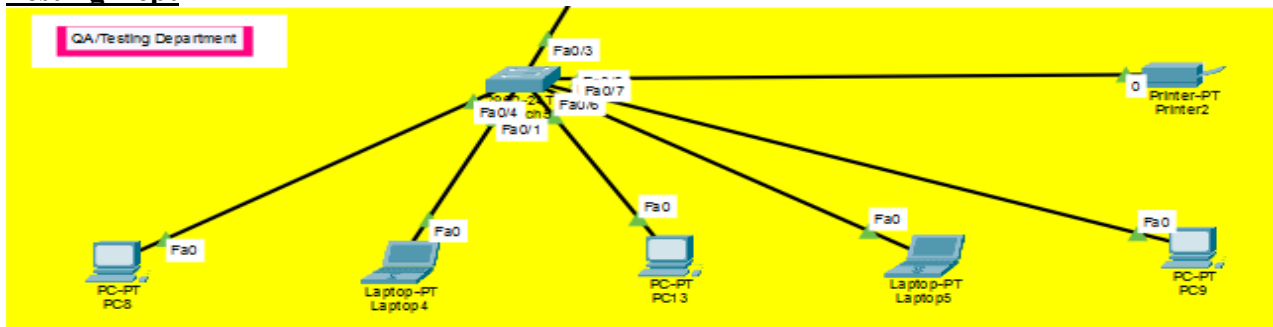
Management Dept



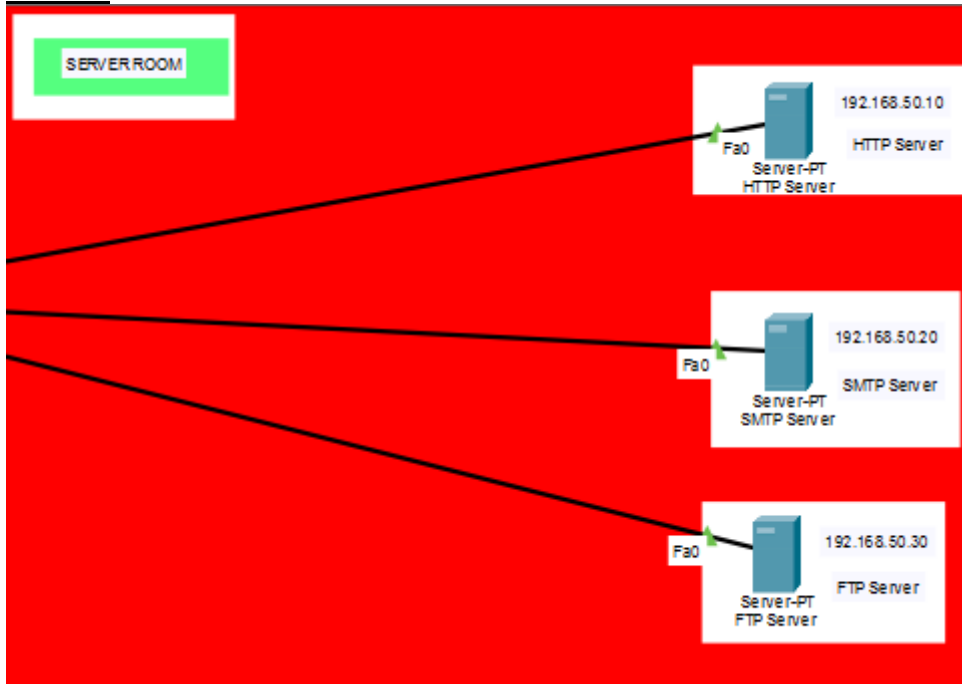
Sales Dept



Testing Dept



Servers



File Uploading (FTP)

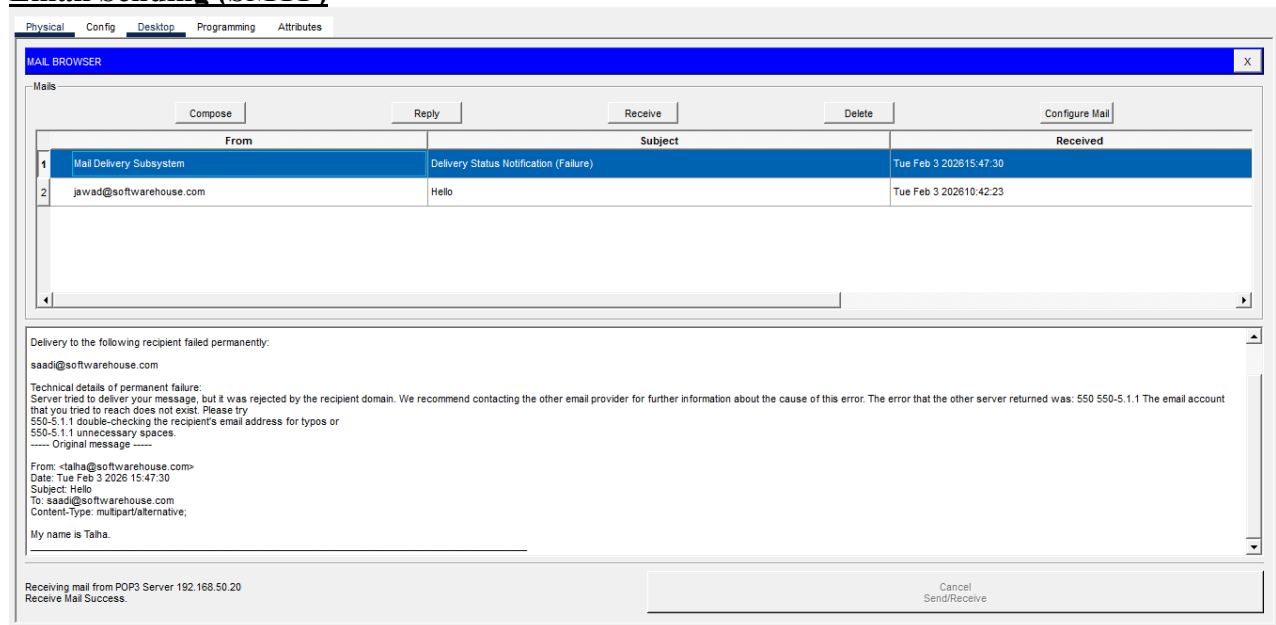
```
Physical  Config  Desktop  Programming  Attributes

Command Prompt

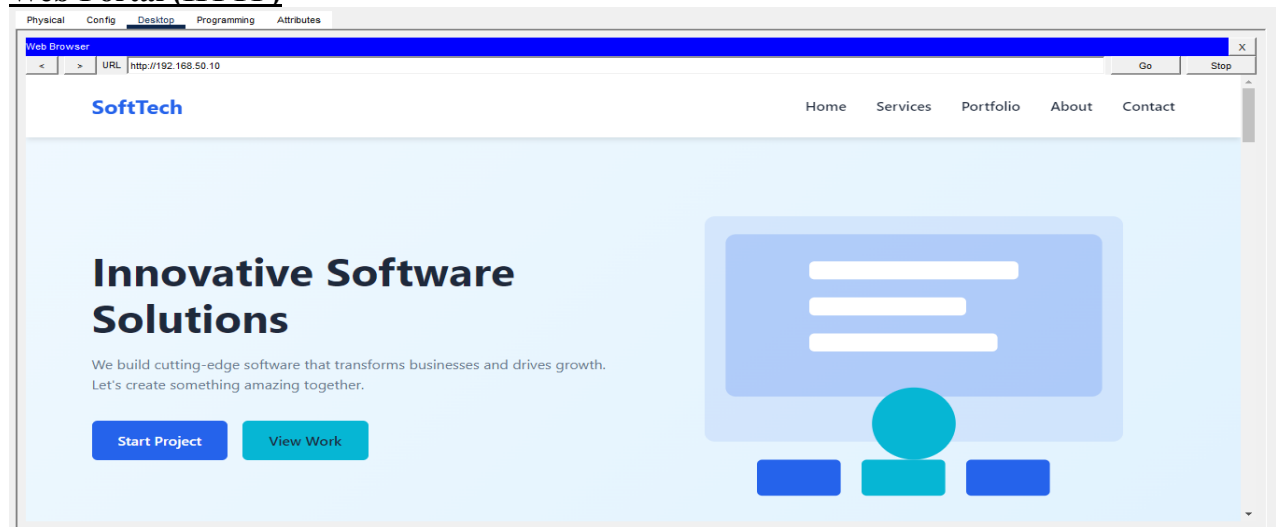
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.50.30
Trying to connect... 192.168.50.30
Connected to 192.168.50.30
220- Welcome to FT Ftp server
Username:talha
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.50.30:
 0 : File22.txt                23
 1 : Jawad.txt                 17
 2 : Rafay.txt                 16
 3 : asa942-k8.bin             5571584
 4 : asa923-k8.bin             30468096
 5 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 6 : c1841-ipbasek9-mz.123-14.T7.bin       13832032
 7 : c1841-ipbasek9-mz.124-12.bin          16599160
 8 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 9 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
10 : c2600-i-mz.122-26.bin               5571584
11 : c2600-ipbasek9-mz.124-8.bin          13169700
12 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
13 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
14 : c2800nm-ipbase-mz.123-14.T7.bin       5571584
15 : c2800nm-ipbasek9-mz.124-8.bin          15522644
16 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
17 : c2960-16q412-mz.121-22.EA4.bin        3058048
18 : c2960-16q412-mz.121-22.EA9.bin        3117390
19 : c2960-lanbase-mz.122-26.FX.bin         4414321
20 : c2960-lanbase-mz.122-26.SE21.bin       4670455
21 : c2960-lanbasek9-mz.150-2.SE4.bin       4670455
22 : c3660-advipservicesk9-mz.122-37.SE1.bin 8662194
23 : c3660-advipservicesk9-mz.122-46.SE.bin 10711279
24 : c800-universalk9-mz.SPA.152-4.M4a.bin  33591768
25 : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
26 : cat3k_caa-universalk9.16.05.02.SPA.bin 505532849
27 : cgr1000-universalk9-mz.SPA.154-2.CG    153487552
28 : cgr1000-universalk9-mz.SPA.156-3.CG    184630138
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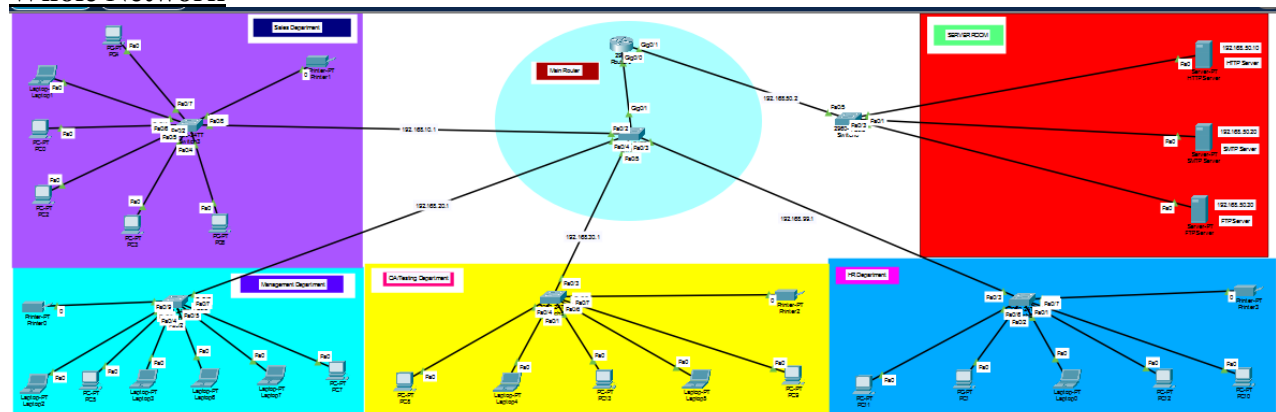
Email Sending (SMTP)



Web Portal (HTTP)



Whole Network



5. Network Design

Topology

A **star topology** is used within departments, connected through switches.

IP Addressing

- Private IP addressing scheme (e.g., 192.168.1.0/24)
- Static IPs for servers

Switching and VLANs

- VLANs are used to logically separate departments
- Inter-VLAN communication is managed through routing

6. Protocols and Technologies Used

- **TCP/IP:** Core communication protocol
- **HTTP:** Web-based services
- **FTP:** File transfer services
- **SMTP & POP3:** Email communication
- **DNS:** Name resolution

These protocols are selected due to their reliability and industry-wide adoption.

7. Implementation Details

The network is implemented using a network simulation tool.

Steps:

1. Network topology creation
2. IP address assignment
3. Switch and router configuration
4. DHCP server configuration
5. Web, FTP, and Mail server setup
6. Testing connectivity and services

8. Performance Evaluation

Network performance is evaluated based on:

- **Connectivity:** Successful ping between devices
- **Service Availability:** Access to HTTP, FTP, and Email services
- **Response Time:** Acceptable latency within LAN

The results show stable and efficient performance.

9. Security Considerations

Basic security measures include:

- Controlled access to servers
- VLAN-based traffic isolation
- Password protection on network devices

These measures help protect the network from unauthorized access.

10. Results and Analysis

The implemented network successfully:

- Assigns IP addresses using static Address
- Allows users to access web and FTP services
- Enables email communication between users

The observed results match the expected outcomes.

11. Discussion

The SoftTech network demonstrates how a properly planned network improves efficiency and manageability. The hierarchical design simplifies troubleshooting and supports future expansion.

12. Challenges and Limitations

- Initial IP addressing issues
- Service configuration errors during setup
- Limited real-hardware testing

13. Applications and Use Cases

- Small to medium software houses
- Office LAN environments
- Educational network simulations

14. Conclusion

This project successfully implements a functional and scalable software house network. It provides hands-on experience with real-world networking concepts, protocols, and services, fulfilling the objectives of the Computer Networks course.

15. Future Enhancements

- Implementation of advanced security
- Wireless network integration
- Network monitoring and logging
- Redundancy and load balancing

16. References

1. Andrew S. Tanenbaum, *Computer Networks*, Pearson Education
2. Cisco Networking Academy – Packet Tracer Documentation